



Aalto University

Moment Estimator-Based Extreme Quantile Estimation with Erroneous Observations

Joint work with Pauliina Ilmonen and Lauri Viitasaari

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Agenda of the Presentation

Motivation

Preliminaries of Extreme Value Theory

Extreme Value Theory with Approximations

Extreme Quantile Region Estimation under Ellipticity

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Extreme Value Theory: Tools for Extrapolation

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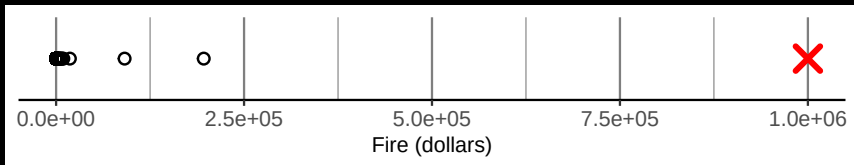


Figure: Insurance company keeps records of fire insurance claims per windstorm. In total, the figure shows 700 observations.

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What if instead of the original observations only approximations are available?

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Why?

- Extreme value index estimation for projections of functional data (Kokoszka and Xiong 2018).
- Extreme quantile/expectile regression (Girard et al. 2021).
- Latent model extreme value index estimation (Virta et al. 2024).

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Much more work has been done especially in the heavy-tailed case (few or no moments exist)...

Approaches from the literature (for the heavy tail)

1. (Kim and Kokoszka 2020; Kim and Kokoszka 2023): We have $\hat{X}_i = X_i + \varepsilon_i$, where

- the errors ε_i are i.i.d.,
- $\{X_i\} \perp\!\!\!\perp \{\varepsilon_i\}$, and
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2. (Girard et al. 2021): We have that the errors $|\hat{X}_i - X_i|$

- can be dependent,
- they do not need to be identically distributed, and
- $\{X_i\} \perp\!\!\!\perp \{\varepsilon_i\}$ can fail.
- “Quantiles of $|\hat{X}_i - X_i|$ grow sufficiently slowly”

Common message

If errors are less heavy-tailed compared to the observations, life is good.

Scope of the Presentation

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1. We quantify the effect of approximation errors for a certain *extreme quantile estimator*.
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2. Application to multivariate extreme quantile region estimation under ellipticity.

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Maximum Domain of Attraction ($Y \in \text{MDA}(G)$)

Let $Y_1, \dots, Y_n$ be i.i.d. observations of a random variable Y . We have $Y \in \text{MDA}(G)$ if there exist sequences $a_n > 0$ and $b_n \in \mathbb{R}$, and a random variable G with a nondegenerate distribution such that

$$\frac{\max(Y_1, \dots, Y_n) - b_n}{a_n} \xrightarrow{\mathcal{D}} G, \quad n \rightarrow \infty.$$

Extreme Value Index (Fisher and Tippett 1928; Gnedenko 1943)

Up to location and scale, the distribution of $G = G_\gamma$ is characterized by the parameter γ , called the *extreme value index*. That is, the distribution of G_γ is of the type

$$F_{G_\gamma}(x) = \begin{cases} \exp\left(-(1 + \gamma x)^{-1/\gamma}\right), & 1 + \gamma x > 0 \quad \text{if } \gamma \neq 0, \\ \exp(-e^{-x}), & x \in \mathbb{R} \quad \text{if } \gamma = 0. \end{cases}$$

Maximum domain of attraction $\text{MDA}(G_\gamma)$ for different values of the extreme value index γ .

γ	Example of $F \in \text{MDA}(G_\gamma)$	Properties of F
> 0	Pareto distribution	For a larger γ fewer moments exist*
$= 0$	Normal distribution	All moments exist*
< 0	Uniform distribution	Finite endpoint

*The claims about the moments hold if the left tail of a distribution is not considered.

Tail Quantile Function

Define the tail quantile function corresponding to a distribution F by

$$U(t) = F^{\leftarrow} \left(1 - \frac{1}{t} \right), \quad t > 1,$$

where we denote the left-continuous inverse of a nondecreasing function by $f^{\leftarrow}(y) = \inf \{x \in \mathbb{R} : f(x) \geq y\}$.

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That is, $U(1/p)$ is the $(1 - p)$ -quantile.

MDA Condition in Terms of Quantiles (de Haan 1984)

Let U be the tail quantile function corresponding to a random variable Y . Then, $Y \in \text{MDA}(G_\gamma)$ if and only if there exists a positive function $a : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that for all $x > 0$,

$$\lim_{t \rightarrow \infty} \frac{U(tx) - U(t)}{a(t)} = \frac{x^\gamma - 1}{\gamma}.$$

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Choose $t = n/k$ and $x = k/(np)$ to get the approximation

$$U\left(\frac{1}{p}\right) \approx U\left(\frac{n}{k}\right) + a\left(\frac{n}{k}\right) \frac{\left(\frac{k}{np}\right)^\gamma - 1}{\gamma}.$$

Extreme Quantile Estimation

Suppose $\mathbf{Y} = (Y_1, \dots, Y_n)$ is an i.i.d. sample of $Y \in \text{MDA}(G_\gamma)$, $\gamma \in \mathbb{R}$. Denote order statistics corresponding to the sample $\mathbf{Y}$ by $\mathbf{Y}_{1,n} \leq \dots \leq \mathbf{Y}_{n,n}$. Then an estimator for the extreme $(1 - p)$ -quantile $x_p = U(1/p)$ can be given as

$$\hat{x}_p(\mathbf{Y}) = \mathbf{Y}_{n-k,n} + \hat{a}\left(\frac{n}{k}\right) \frac{\left(\frac{k}{np}\right)^{\hat{\gamma}(\mathbf{Y})} - 1}{\hat{\gamma}(\mathbf{Y})},$$

where $\hat{\gamma}$ is an estimator for the extreme value index γ .

The Moment Estimator (Dekkers et al. 1989)

Set, for $\ell \in \{1, 2\}$,

$$M_n^{(\ell)}(\mathbf{Y}) = \frac{1}{k} \sum_{j=0}^{k-1} (\ln \mathbf{Y}_{n-j,n} - \ln \mathbf{Y}_{n-k,n})^\ell,$$

and

$$\hat{\gamma}_+(\mathbf{Y}) = M_n^{(1)}(\mathbf{Y}) \quad \text{and} \quad \hat{\gamma}_-(\mathbf{Y}) = 1 - \frac{1}{2} \left(1 - \frac{(M_n^{(1)}(\mathbf{Y}))^2}{M_n^{(2)}(\mathbf{Y})} \right)^{-1}.$$

Then the moment estimator is defined as

$$\hat{\gamma}_M(\mathbf{Y}) = \hat{\gamma}_+(\mathbf{Y}) + \hat{\gamma}_-(\mathbf{Y}),$$

and the corresponding estimator for $a(n/k)$ is defined as

$$\hat{\sigma}_M(\mathbf{Y}) = \mathbf{Y}_{n-k,n} M_n^{(1)}(\mathbf{Y}) (1 - \hat{\gamma}_-(\mathbf{Y})).$$

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Back to Approximations

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- How the approximation error affects the asymptotics?

That is, next we quantify how the error

$$\max_{0 \leq j \leq k} \left| \frac{\hat{Y}_{n-j,n}}{Y_{n-j,n}} - 1 \right|$$

has to behave in order to have asymptotic normality for $\hat{\chi}_\rho(\hat{\mathbf{Y}})$.

Assumptions for the Main Result

1. Y is an almost surely positive random variable with $F \in \mathcal{D}(G_\gamma)$, $\gamma \in \mathbb{R}$ (and U satisfies certain second-order conditions).
2. $\mathbf{Y} = (Y_1, \dots, Y_n)$ are i.i.d. copies of Y and let $\hat{\mathbf{Y}} = (\hat{Y}_1, \dots, \hat{Y}_n)$ be an approximated sample.

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3. Suppose $k = k_n \rightarrow \infty$, $k_n/n \rightarrow 0$, as $n \rightarrow \infty$.
4. $p = p_n$, $np_n = o(k_n)$ and $\ln(np_n) = o(\sqrt{k_n})$.

Sketch of the Main Result

Suppose that assumptions of the previous slide hold and that

$$\max_{0 \leq j \leq k_n} \left| \frac{\hat{\mathbf{Y}}_{n-j,n}}{\mathbf{Y}_{n-j,n}} - 1 \right| = O_{\mathbb{P}}(h_n)$$

for some sequence h_n such that $\sqrt{k_n} \frac{h_n U\left(\frac{n}{k_n}\right)}{a\left(\frac{n}{k_n}\right)} = o(1)$. Denote

$q_{\gamma}(t) = \int_1^t s^{\gamma-1} \ln s \, ds$. Then

$$\sqrt{k_n} \frac{\hat{x}_{p_n}(\hat{\mathbf{Y}}) - U(1/p_n)}{a\left(\frac{n}{k_n}\right) q_{\gamma}\left(\frac{k_n}{np_n}\right)}$$

is asymptotically normal.

Condition for the Error

Suppose $k = k_n \rightarrow \infty$, $k_n/n \rightarrow 0$, as $n \rightarrow \infty$. Assume also that U satisfies a certain second-order condition and that

$$\max_{0 \leq j \leq k_n} \left| \frac{\hat{\mathbf{Y}}_{n-j,n}}{\mathbf{Y}_{n-j,n}} - 1 \right| = O_{\mathbb{P}}(h_n)$$

for some sequence h_n . Then $\sqrt{k_n} \frac{h_n U\left(\frac{n}{k_n}\right)}{a\left(\frac{n}{k_n}\right)} \rightarrow 0$ if

1. $\gamma > 0$ and $\sqrt{k_n} h_n \rightarrow 0$,
2. $\gamma = 0$ and $\sqrt{k_n} h_n = O\left(\left(\frac{n}{k_n}\right)^{-\delta}\right)$ for some $\delta > 0$,
3. $\gamma < 0$ and $\sqrt{k_n} h_n = o\left(\left(\frac{n}{k_n}\right)^{\gamma}\right)$.

Example

If

$$\max_{0 \leq j \leq k_n} \left| \frac{\hat{\mathbf{y}}_{n-j,n}}{\mathbf{y}_{n-j,n}} - 1 \right| = o_{\mathbb{P}} \left(\frac{1}{\sqrt{n}} \right),$$

then the sufficient condition for the error does not hold if $\gamma \leq -1/2$!

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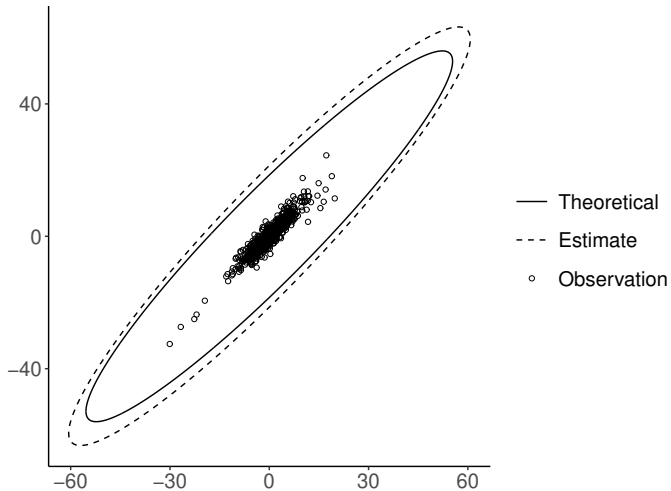
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t-distribution, $n = 1000$, $k = 100$, $\gamma = 0.25$, $p = 1/5000$, $\alpha_{\text{MCD}} = 0.5$



Extreme Quantile Region Estimation

Let $X \in \mathbb{R}^d$ be a random vector with density f . We wish to estimate

$$Q_p = \{x \in \mathbb{R}^d : f(x) \leq \beta_p\},$$

where β_p is chosen such that $\mathbb{P}(X \in Q_p) = p$, and p is small.

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The problem has been considered by

- Cai et al. 2011, and
- He and Einmahl 2017 in the context of halfspace depth contours under multivariate regular variation.

Elliptical Distributions (Cambanis et al. 1981)

We say that a d -variate random vector X is elliptically distributed if

$$X \stackrel{\mathcal{D}}{=} \mu + \mathcal{R}\Lambda S,$$

where

- $\mu \in \mathbb{R}^d$ is called the *location vector*,
- $\mathcal{R}$ is a nonnegative random variable called the *generating variate*,
- $\Lambda \in \mathbb{R}^{d \times d}$ is a matrix such that $\Sigma = \Lambda\Lambda^\top$ is a symmetric positive definite matrix (matrix Σ is called *scatter matrix*),
- S is an d -dimensional random vector uniformly distributed over the unit-sphere $\{x \in \mathbb{R}^d : x^\top x = 1\}$ and
- Random variables $\mathcal{R}$ and S are independent.

Role of the Generating Variate $\mathcal{R}$

- We suppose $\mathcal{R} \in \text{MDA}(G_\gamma)$ for $\gamma \in \mathbb{R}$

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Role of the Generating Variate $\mathcal{R}$

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- $\mathcal{R} \stackrel{\mathcal{D}}{=} \sqrt{(X - \mu)^\top \Sigma^{-1} (X - \mu)}$
- $\hat{R}_i = \sqrt{(X_i - \hat{\mu}(\mathbf{X}))^\top \hat{\Sigma}^{-1}(\mathbf{X}) (X_i - \hat{\mu}(\mathbf{X}))}$
- $\hat{\mathbf{R}} = (\hat{R}_1, \dots, \hat{R}_n)$

The Estimator

- We wish to estimate

$$Q_p = \left\{ x \in \mathbb{R}^d : \sqrt{(x - \mu)^\top \Sigma^{-1} (x - \mu)} \geq r_p \right\},$$

for a very small p , where r_p is the $(1 - p)$ -quantile of the generating variate $\mathcal{R}$.

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- Define estimator $\hat{Q}_p$ as

$$\hat{Q}_p = \left\{ x \in \mathbb{R}^d : \sqrt{(x - \hat{\mu}(\mathbf{X}))^\top \hat{\Sigma}^{-1}(\mathbf{X}) (x - \hat{\mu}(\mathbf{X}))} \geq \hat{x}_p(\hat{\mathbf{R}}) \right\},$$

where $\hat{x}_p(\hat{\mathbf{R}})$ is the extreme quantile estimator computed with approximations $\hat{\mathbf{R}}$.

Approximation Error for the Generating Variate

Assume $k = k_n \rightarrow \infty$, $k_n/n \rightarrow 0$, as $n \rightarrow \infty$ and that estimators of the location and scatter satisfy $\sqrt{n}(\hat{\mu}_n - \mu) = O_{\mathbb{P}}(1)$ and $\sqrt{n}(\hat{\Sigma}_n - \Sigma) = O_{\mathbb{P}}(1)$. Then

$$\max_{0 \leq j \leq k_n} \left| \frac{\hat{\mathbf{R}}_{n-j,n}}{\mathbf{R}_{n-j,n}} - 1 \right| = O_{\mathbb{P}}\left(\frac{1}{\sqrt{n}}\right).$$

Consistency and Affine Equivariance

- **Consistency:** if estimators of the location and scatter are $\sqrt{n}$ -consistent, $\mathcal{R} \in \text{MDA}(\mathbf{G}_\gamma)$ for $\gamma > -1/2$, , $p_n \rightarrow 0$ with a suitable rate, as $n \rightarrow \infty$ (and other technical conditions),

$$\frac{\mathbb{P}\left(X \in \hat{Q}_{p_n} \triangle Q_{p_n}\right)}{p_n} \rightarrow 0, \quad \text{as } n \rightarrow \infty,$$

where $B_1 \triangle B_2 = (B_1 \setminus B_2) \cup (B_2 \setminus B_1)$, $B_1, B_2 \in \mathbb{R}^d$.

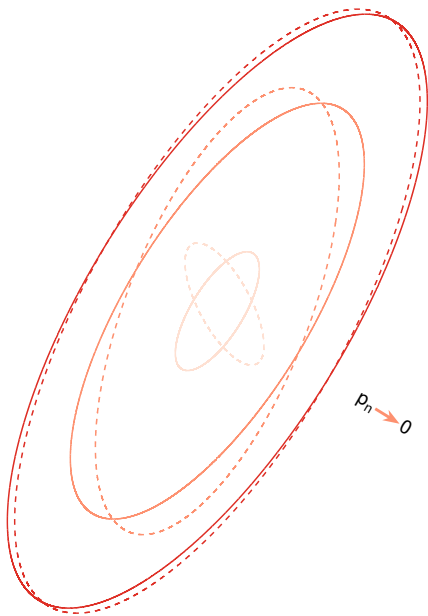
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- **Equivariance under affine transformations** $Ax + b$, $A \in \mathbb{R}^{d \times d}$ invertible.



— Theoretical

- - - Estimate

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